$\equiv \exists \varepsilon > 0 \ \forall \delta > 0 \ \exists x\, ((0 < |x-a| < \delta) \wedge (|f(x) - L| \geq \varepsilon))$

Because the statement "$\lim_{x \to a} f(x)$ does not exist" means for all real numbers L, $\lim_{x \to a} f(x) \neq L$, this can be expressed as

$$\boxed{\forall L \ \exists \varepsilon > 0 \ \forall \delta > 0 \ \exists x\, (0 < |x-a| < \delta \wedge |f(x) - L| \geq \varepsilon)}.$$

This last statement says that for every real number L there is a real number $\varepsilon > 0$ such that for every real number $\delta > 0$, there exists a real number x such that $0 < |x-a| < \delta$ and $|f(x) - L| \geq \varepsilon$.